

Exercise 1

以连杆 1 关节轴为基坐标和重力参考零势面

广义坐标参数: $\mathbf{q} = \begin{pmatrix} \theta_1 \\ d_2 \end{pmatrix}$

动能:

$$\text{连杆 1 质心位置} \begin{cases} x_{c_1} = l_1 \cos \theta_1 \\ y_{c_1} = l_1 \sin \theta_1 \end{cases}$$

$$\text{质心速度} \begin{cases} v_{x,c_1} = -l_1 \sin \theta_1 \dot{\theta}_1 \\ v_{y,c_1} = l_1 \cos \theta_1 \dot{\theta}_1 \end{cases}, \quad v_{c_1} = l_1 \dot{\theta}_1$$

连杆 1 质心运动与绕质心转动结合 $K_1 = \frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2}I_1 \dot{\theta}_1^2$

$$\text{连杆 2 质心位置} \begin{cases} x_{c_2} = d_2 \cos \theta_1 \\ y_{c_2} = d_2 \sin \theta_1 \end{cases}$$

$$\text{质心速度} \begin{cases} v_{x,c_2} = \dot{d}_2 \cos \theta_1 - d_2 \sin \theta_1 \dot{\theta}_1 \\ +v_{y,c_2} = \dot{d}_2 \sin \theta_1 + d_2 \cos \theta_1 \dot{\theta}_1 \end{cases}$$

$$v_{c_2} = \sqrt{v_{x,c_2}^2 + v_{y,c_2}^2} = \sqrt{\dot{d}_2^2 + d_2^2 \dot{\theta}_1^2}$$

连杆 2 质心运动与绕质心转动结合 $K_2 = \frac{1}{2}m_2(\dot{d}_2^2 + d_2^2 \dot{\theta}_1^2) + \frac{1}{2}I_2 \dot{\theta}_1^2$

动能

$$K = K_1 + K_2 = \frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2(\dot{d}_2^2 + d_2^2 \dot{\theta}_1^2) + \frac{1}{2}(I_1 + I_2)\dot{\theta}_1^2$$

势能:

$$U = m_1 g l_1 \sin \theta_1 + m_2 g d_2 \sin \theta_1$$

计算偏导数

$$\begin{aligned}\frac{\partial L}{\partial \dot{\theta}_1} &= m_1 l_1^2 \dot{\theta}_1 + m_2 d_2^2 \dot{\theta}_1 + (I_1 + I_2) \dot{\theta}_1 \\ \frac{\partial L}{\partial \dot{d}_2} &= m_2 \dot{d}_2 \\ \frac{\partial L}{\partial \theta_1} &= -m_1 g l_1 \cos \theta_1 - m_2 g d_2 \cos \theta_1 \\ \frac{\partial L}{\partial d_2} &= m_2 d_2 \dot{\theta}_1^2 - m_2 g \sin \theta_1\end{aligned}$$

广义力:

设连杆 1 电机扭矩为 τ_1 , 连杆 2 电机力为 f_2 , $\mathbf{F} = \begin{pmatrix} \tau_1 \\ f_2 \end{pmatrix}$

应用拉格朗日方程得到动力学:

$$\begin{aligned}\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} &= m_1 l_1^2 \ddot{\theta}_1 + 2m_2 d_2 \dot{\theta}_1 \dot{d}_2 + m_2 d_2^2 \ddot{\theta}_1 + (I_1 + I_2) \ddot{\theta}_1 \\ \frac{d}{dt} \frac{\partial L}{\partial \dot{d}_2} &= m_2 \ddot{d}_2 \\ &\begin{pmatrix} m_1 l_1^2 + m_2 d_2^2 + I_1 + I_2 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} \ddot{\theta}_1 \\ \ddot{d}_2 \end{pmatrix} \\ &\quad + \begin{pmatrix} 2m_2 d_2 \dot{\theta}_1 \dot{d}_2 \\ -m_2 d_2 \dot{\theta}_1^2 \end{pmatrix} \\ &\quad + \begin{pmatrix} m_1 g l_1 \cos \theta_1 + m_2 g d_2 \cos \theta_1 \\ m_2 g \sin \theta_1 \end{pmatrix} \\ &= \begin{pmatrix} \tau_1 \\ f_2 \end{pmatrix}\end{aligned}$$

Exercise 2

如下图

连杆 1 的质量为 m_1 , 长度为 l_1 , 质心离底面的距离为 d_1 , 连杆 2

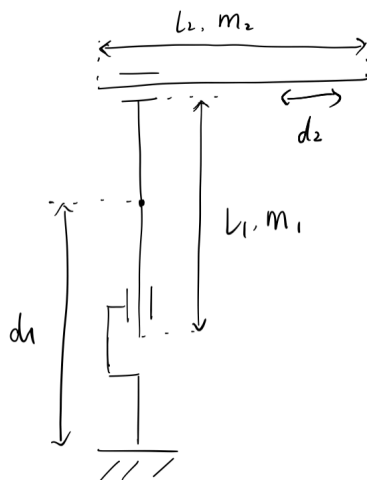


图 1: exercise 2

的质量为 m_2 ，将底面作为重力零势面

$$\text{连杆 1 质心速度} \begin{cases} v_{x,c_1} = 0 \\ v_{y,c_1} = \dot{d}_1 \end{cases}, \quad v_{c_1} = \dot{d}_1$$

$$\text{连杆 1 动能 } K_1 = \frac{1}{2} m_1 \dot{d}_1^2$$

$$\text{连杆 2 质心速度} \begin{cases} v_{x,c_2} = \dot{d}_2 \\ v_{y,c_2} = \dot{d}_1 \end{cases}, \quad v_{c_2} = \sqrt{\dot{d}_1^2 + \dot{d}_2^2}$$

$$\text{连杆 2 动能 } K_2 = \frac{1}{2} m_2 (\dot{d}_1^2 + \dot{d}_2^2)$$

动能:

$$K = K_1 + K_2 = \frac{1}{2} (m_1 + m_2) \dot{d}_1^2 + \frac{1}{2} m_2 \dot{d}_2^2$$

势能

$$U = m_1 g d_1 + m_2 g (d_1 + \frac{1}{2} l_1)$$

计算偏导数

$$\begin{aligned}\frac{\partial L}{\partial \dot{d}_1} &= (m_1 + m_2)\dot{d}_1 \\ \frac{\partial L}{\partial \dot{d}_2} &= m_2\dot{d}_2 \\ \frac{\partial L}{\partial d_1} &= -m_1g - m_2g \\ \frac{\partial L}{\partial d_2} &= 0\end{aligned}$$

广义力: 设连杆 1 电机力为 f_1 , 连杆 2 电机力为 f_2 , $\mathbf{F} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$

应用拉格朗日方程得到动力学:

$$\begin{aligned}\frac{d}{dt} \frac{\partial L}{\partial \dot{d}_1} &= (m_1 + m_2)\ddot{d}_1 \\ \frac{d}{dt} \frac{\partial L}{\partial \dot{d}_2} &= m_2\ddot{d}_2\end{aligned}$$

$$\begin{pmatrix} m_1 + m_2 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} \ddot{d}_1 \\ \ddot{d}_2 \end{pmatrix} + \begin{pmatrix} (m_1 + m_2)g \\ 0 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$